Hospital Delivery Robot Planner

Innovative Assignment II — Foundations of Artificial Intelligence

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1. Problem Formulation

Phase 1 solves a grid maze from a start cell to a goal cell. A state is a (row, column) position; actions move one cell up, down, left or right; walls are impassable; each move costs 1. The goal test checks whether the current cell equals the goal.

2. Approach

BFS uses a FIFO queue and explores cells in increasing path distance. A* selects the frontier state with minimum f(n)=g(n)+h(n), where g is the path cost so far and h is Manhattan distance to the goal. For four-direction movement with unit costs, Manhattan distance is admissible. The extension schedules deliveries as a CSP: tasks are variables, time slots are domains, and shared-robot tasks have inequality constraints. AC-3 prunes unsupported values; MRV chooses the next variable; backtracking finds a consistent assignment.

3. Hospital Extension

After scheduling, a rule-based layer marks busy or blocked grid cells as unavailable and triggers A* replanning. Explanations identify the cells avoided and note the sterile-robot requirement for ICU delivery. If no path exists, the program reports that human intervention is required.

4. Complexity

For a graph with V traversable cells and E neighbour connections, BFS is O(V+E) time and O(V) space. A* worst-case time and space are exponential in search depth in general; on this finite grid, it is bounded by the number of reachable cells for this implementation with a consistent heuristic and duplicate checks. AC-3 worst-case complexity is commonly O(e d^3), where e is the number of constraint arcs and d is the maximum domain size; backtracking remains exponential in the number of tasks in the worst case.

5. Results

Run `python src/main.py` to generate the actual path lengths and expanded-node counts for the supplied maze, along with the CSP schedule and dynamic-obstacle route. The run output is saved in docs/sample_output.txt. Replace or supplement this text with screenshots from your own run before submission.

6. Reflection and Responsible AI

The project demonstrates the difference between uninformed and heuristic search and connects route planning to scheduling constraints. Current limitations include manually entered obstacles, a small synthetic maze, and simplified hospital rules. A real system would need validated sensor inputs, uncertainty calibration, privacy protections, sterile-equipment verification, human oversight, and safety testing before deployment. This is an educational prototype and must not control real medical deliveries.

Sample execution output

```
=== PHASE 1: MAZE SOLVER ===
Original maze (# wall, . free, S start, G goal):
S . . . . . .
. # # # # # .
. . . . . # .
. # # # . # .
. . . . . . .
. # # # # # .
```

```
. . . . . G

BFS result
S * * * * * *
. # # # # # *
. . . . . # *
. # # # . # *
. . . . . . *
. # # # # # *
. . . . . G
Path found: True | Path steps: 12 | Nodes expanded: 33

A* result
S * * * * * *
. # # # # # *
. . . . . # *
. # # # . # *
. . . . . . *
. # # # # # *
. . . . . G
Path found: True | Path steps: 12 | Nodes expanded: 33

=== PHASE 2: HOSPITAL DELIVERY PLANNER ===
Delivery schedule (CSP + MRV + AC-3):
{'Medicine-ICU': '09:00', 'Sample-Lab': '10:00', 'Medicine-Ward': '11:00'}

Route with dynamic obstacle:
S * * * * * *
. # # # # # *
. . # . . # *
. # # # . # *
. . . . . . *
. # # # # # *
. . . . . G
Path steps: 12 | Nodes expanded: 29
Explanation:
- Rule fired: cell (2, 2) is busy/blocked; route planner avoids it.
- Safety rule: deliveries entering ICU zones require a sterile robot.
```

Honesty note: AI assistance may have been used to scaffold code and documentation. The student should understand, personalise, test, and disclose assistance according to course rules.